

Read each question completely before answering. Each question is worth 2 points except as noted.

Multiple choice. Choose the *best* answer. Clearly circle the letter for your answer. Vague or unclear marks will be counted as incorrect.

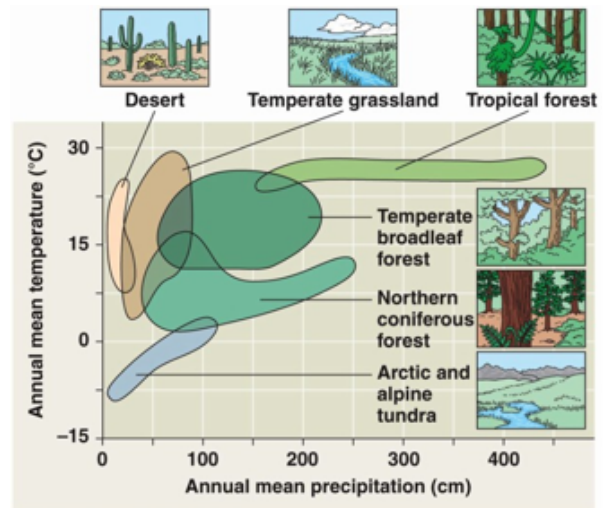
1. Symbiosis is a long-term ecological interaction
 - A. where one species benefits and another species is harmed
 - B. where two or more species partition resources.
 - C. between two or more species.**
 - D. where one species benefits and the other species neither benefits nor is harmed.
 - E. where both species benefit.
 2. Early successional species add nutrients and deepen the soil, allowing intermediate successional species to colonize an area. This process is known as
 - A. inhibition.
 - B. primary succession.
 - C. tolerance.
 - D. disturbance.
 - E. facilitation.**
 - F. secondary succession.
 3. One form of elephantiasis occurs when *Brugia malayyi*, a thread-like worm, invades human lymph nodes. The worm blocks the flow of lymph throughout the body, causing massive swelling of the lower body, especially legs and genitals. As described here, this is an example of
 - A. symbiosis.
 - B. parasitism.**
 - C. mutualism.
 - D. commensalism.
 - E. predation.
 4. Energy from the sun is converted into chemical energy for organisms by
 - A. primary consumers.
 - B. primary producers.**
 - C. herbivores.
 - D. decomposers.
 - E. carnivores.
 5. Which dispersion should be favored among plants that compete intraspecifically by casting shade?
 - A. Clumped.
 - B. Clustered.
 - C. Evenly spaced.**
 - D. Random.
-

6. Which statement most accurately describes how nutrients and energy are used in ecosystems?

- A. Nutrients can be converted into energy; energy cannot be converted into nutrients.
- B. Energy can be converted into nutrients; nutrients cannot be converted into energy.
- C. Energy flows through ecosystems; nutrients cycle within ecosystems.**
- D. Nutrients flow through ecosystems; energy cycles within ecosystems.
- E. Nutrients are used in ecosystems; energy is not.

7. According to this climograph, the temperate grassland

- A. has an average annual temperature range of about 4–30°C.**
- B. has an average annual rainfall of about 5–30 cm.
- C. has the greatest variation in precipitation of any ecosystem shown in the figure.
- D. never occurs with deserts.
- E. has an average annual rainfall of about 20–80 cm.

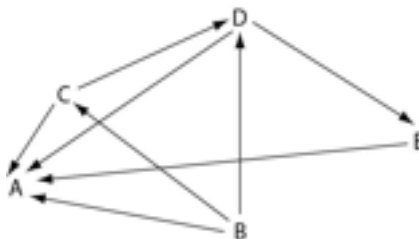


8. Plants produce water vapor during photosynthesis, which goes from the plant to the atmosphere. The water vapor in the atmosphere eventually falls back to the earth as rain. This process would be part of

- A. the nutrient cycle.
- B. the energy cycle.
- C. the water cycle.**
- D. the entire ecosystem.
- E. plant metabolism.
- F. energy flow.

9. Based on the hypothetical food web below, one letter represents an organism that could be both a secondary and tertiary consumer? The arrows point in the direction of energy transfer from lower to higher trophic levels. More than one answer is possible. Circle only one answer.

- A. A**
- B. B
- C. C
- D. D
- E. E

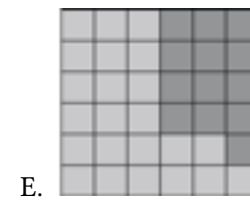
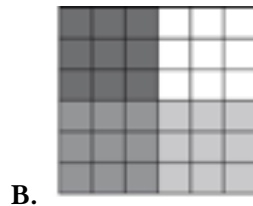
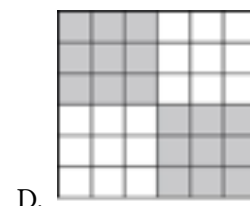
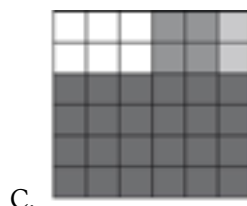
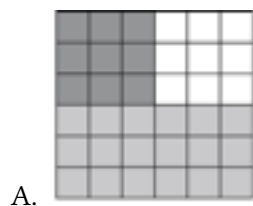


10. Fire suppression by humans
- A. is a management goal of conservation biologists to maintain the healthy condition of forest communities.
 - B. will always result in an increase in species diversity in a given biome.
 - C. can change the species composition within biological communities.**
 - D. is necessary for the protection of threatened and endangered forest species.
 - E. will result ultimately in sustainable production of increased amounts of forest products for human use.
11. The “role” or “job” of a species in a community, represented by all of the resources needed by the species, is called
- A. a predator or prey.
 - B. the resource.
 - C. the niche.**
 - D. a producer or consumer.
 - E. a predator.
12. The energy that is *not* transferred to other trophic levels or decomposers is lost
- A. to consumers at any trophic level.
 - B. to primary consumers.
 - C. in fecal matter.
 - D. as metabolic heat.**
 - E. to secondary consumers.
13. Organisms that break down organic material and make it available to other organisms are called
- A. decomposers.**
 - B. scavengers.
 - C. consumers.
 - D. producers.
 - E. degraders.
14. Which of the following statements about ecosystems is false?
- A. Energy flows through an ecosystem.
 - B. Water cycles within an ecosystem.
 - C. About 10% of energy available at one trophic level is lost as it moves to the next higher trophic level.**
 - D. Carnivores occupy higher trophic levels than herbivores.
 - E. Nutrients, like nitrogen and phosphorus, cycle within an ecosystem.
15. An organism that only reproduces once is known as
- A. perennial.
 - B. iteroparous.
 - C. ectotherm.
 - D. semelparous.**
-

16. A field containing 2000 kg of plant material can support about _____ of tertiary consumers.

- A. **2 kg**
- B. 10%
- C. 10 kg
- D. 20 kg
- E. 90%
- F. 200 kg

17. Which block in the figure below shows the greatest species diversity? (Different shades, including white, represent different species. The small squares indicate relative abundance of individuals present.)



18. Colors and patterns that helps an animal blend in with its environment is called

- A. aposomatic coloration.
- B. warning coloration
- C. environmental mimicry.
- D. habitat mimicry.
- E. **cryptic coloration.**

19. Competitive exclusion means that

- A. **one species will always out complete another species that occupies the same niche.**
- B. two species will partition resources to eliminate competition.
- C. one population will always exclude another another population of the same species in a community.
- D. a species will have a smaller realized niche than a fundamental niche.
- E. one individual will always out complete another individual in a population.

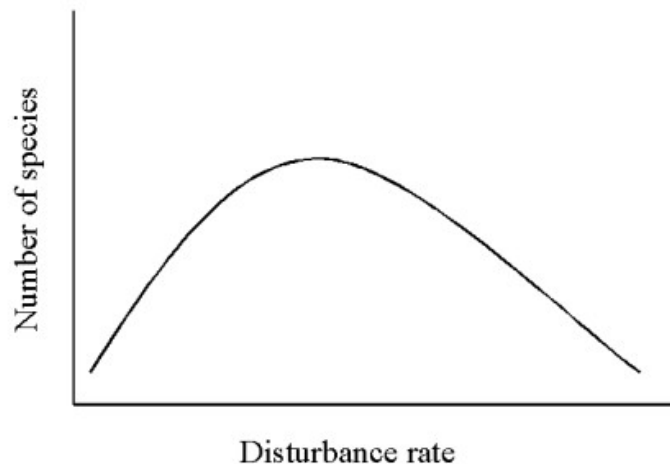
20. In a population growing according to the exponential growth model, population size is

- A. limited by both density-dependent and density-independent factors.
- B. limited by density-dependent factors.
- C. limited by density-independent factors.
- D. **not limited.**

21. Of the following choices, which is the biggest concern about climate change?
- A. The amount of solar radiation will change faster than plants can adapt.
 - B. Forest and grassland ecosystems will die due to less CO₂ available for photosynthesis.
 - C. More regions around the globe will experience stronger rain shadows.
 - D. More regions around the globe will experience desert-like conditions.
 - E. The average global temperature will be warmer than ever recorded.
 - F. The rate of temperature increase is faster than ever recorded.**
22. What type of survivorship curve describes a species that has a high rate of juvenile mortality but low adult mortality?
- A. Type I curve.
 - B. Type II curve.
 - C. Type III curve.**
 - D. Type IV curve
23. The difference between communities and ecosystems is that
- A. communities include individuals of the same species; ecosystems include individuals of different species.
 - B. communities include interactions between individuals and their physical environment; ecosystems do *not* include such interactions.
 - C. communities are groups of individuals of the different species; ecosystems are groups of communities.
 - D. communities include individuals of the different species; ecosystems are individuals of the same species.
 - E. communities do *not* include interactions between individuals and the physical environment; ecosystems *do* include such interactions.**
24. Probably the most important factor(s) affecting the global distribution of ecosystems is (are)
- A. day length and rainfall.
 - B. wind and ocean water current patterns.
 - C. climate.**
 - D. wind direction and rain shadows.
 - E. longitude and latitude.
25. White-breasted nuthatches and brown creepers both eat insects that hide in the furrows of bark in hardwood trees. The brown creeper searches for insects by hunting from the bottom of the tree trunk toward the top, whereas the white-breasted nuthatch searches from the top of the trunk down. Their feeding areas rarely overlap on the tree trunk. These hunting behaviors best illustrate which of the following ecological concepts?
- A. competitive exclusion
 - B. resource partitioning**
 - C. intraspecific competition
 - D. keystone species
 - E. interspecific competition
-

Circle True or False. Read each statement completely before answering.

26. **TRUE** False Collared lemmings self-regulate their populations by mass suicide.
27. True **FALSE** Competitive exclusion will occur when two species use even one identical resource out of many resources.
28. **TRUE** False The brightly colored pattern of poison dart frogs is an example of aposomatic coloration.
29. **TRUE** False Any one trophic level supports on average about 10% of the biomass of the next lower trophic level.
30. **TRUE** False An insect that has evolved to resemble a plant twig will have a better chance of avoiding predation.
31. True **FALSE** Disease is not a factor that could regulate biological populations.
32. True **FALSE** A sink population has more births than deaths.
33. True **FALSE** By adding eggs to an American Kestrel's nest, ecologists found that adult survival increased.
34. True **FALSE** Richness measures only the number of different species in a population.
35. True **FALSE** A field containing 30,000 kg of plant biomass will support about 300 kg of tertiary consumer biomass.
36. (4 points) Draw and properly label a graph that illustrates the intermediate disturbance hypothesis. Each axis must be labeled with what it represents but you do not need to put number scales on them. I will assume low numbers are to left (x) or bottom (y) of an axis and high numbers to the right (x) or top (y) of an axis.



Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice.

Bears	Herbivore	Primary consumer
Carrying capacity	Interspecific competition	Primary producer
Climate change	Iteroparous	Primary production
Community	Keystone species	Primary succession
Commensalism	Mountain lions	Realized Niche
Ecosystem	Mutualism	Secondary consumer
Exploitation	Net primary production	Secondary succession
Global warming	Parasitism	Semelparous
Greenhouse effect	Potential niche	Symbiosis
Gross primary production	Predator	Wolves

37. **Commensalism** One species get ecological benefit over the long term, the other neither benefits or loses.
38. **Secondary consumer** Eaten by tertiary consumers.
39. **Carrying capacity** This is the key distinction between the exponential and logistic population growth models.
40. **Gross primary production** Total amount of sun's energy converted to organic matter.
41. **Primary succession** Following disturbance, begins from bare rock.
42. **Wolves** Hypothesized to cause population fluctuations of moose on Isle Royale
43. **Ecosystem** Interaction between a community and the physical environment.
44. **Realized niche** Based on the resources actually used by a species.
45. **Iteroparous** An individual that is a repeat reproducer is known as this.
46. **Keystone species** A species that greatly increases community diversity.
47. **Greenhouse Effect** Carbon dioxide, methane, and other gasses provide insulation for Earth.

48. (4 points) In population ecology, what does BIDE stand for?

Births, immigration, deaths, emigration. 1 point per term.